

Aziz Mohammad

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EDUCATION

University of Illinois Chicago, Chicago, IL

Aug 2025 – May 2027 (Expected)

BS in Mechanical Engineering

GPA – 3.6/4.0 | Technical GPA – 3.77/4.0 | Dean's List

Relevant Coursework: Fluid Mechanics, Thermodynamics 1&2

College of DuPage, Glen Ellyn, IL

Aug 2023 – May 2025

WORK EXPERIENCE

Wilo USA, Systems Engineering Intern, Rosemont, IL

May 2026 – August 2026

- Spearheaded the ground-up mechanical design and platform architecture for a modular family of commercial HVAC hydronic pump packages (30-230 tons) converting Custom one-off skid requests into 20+ standardized configurations to establish a repeatable product line.
- Coordinated mechanical, hydraulic, and electrical subsystems into a unified, skid-mounted frame, modeling 5" grooved fluid piping loops, specifying thermal pipe insulation, and routing electrical conduit alongside dual-pump setups, VFD string codes, expansion vessels, thermal buffer storage, air/dirt separators, and single-point power entries.
- Built structured configuration matrices, automated Excel costing models, and standardized hardware datasets across international DIN/US standards, eliminating redundant drafting tasks and significantly accelerating sales quotation response times.

University of Illinois at Chicago, Chicago, IL

March 2026 – Present

Research Assistant | Intelligent Multiphase Systems Lab

- Developing a precision flow-boiling experimental apparatus to investigate multiphase heat transfer and interfacial transport physics; utilizing SolidWorks for complex CAD modeling and advanced fabrication to ensure high-fidelity data acquisition for electronics cooling.
- Designed and fabricated custom heater insulation for vapor chamber testing using SLA 3D printing; optimized resin-based prints to meet precise thermal and structural requirements for laboratory experimentation.

CLUBS AND ACTIVITIES

Society of Automotive Engineers (SAE), Member

Aug 2025 – Present

- Optimized a custom FSAE push bar using SolidWorks and Ansys FEA, achieving a **20%** weight reduction while maintaining a **2.5x** factor of safety to enhance vehicle agility and meet stringent competition structural requirements.
- Engineered and fabricated a precision welding jig for the vehicle's tubular space frame, maintaining **±1.5 mm** tolerances to ensure perfect chassis symmetry and the alignment of critical suspension hard points.

American Society of Mechanical Engineers (ASME), Member

Aug 2025 – Present

- Collaborated on cross-disciplinary engineering teams to develop hardware solutions using mechanical principles and CAD modeling, contributing to technical design reviews and peer-reviewed projects focused on industry-standard manufacturing processes.

LEADERSHIP EXPERIENCE

Muslim Student Association (MSA), Member/Volunteer

Aug 2023 – Present

- Coordinated community service initiatives and volunteer events by collaborating with peers to foster engagement while facilitating regular meetings to discuss recruitment strategies and organizational growth.

SKILLS

- **Software:** SolidWorks, Creo, AutoCAD, Ansys Workbench, MATLAB, Python, Microsoft Office, Google Workspace
- **Hardware:** 3D Printing, CNC, Manual Machining, GD&T